The Kakeya Conjecture over Finite Fields and Pandrosion Arithmetic

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Abstract

We develop the Pandrosion framework over finite fields $\mathbb{F}_q$. The fundamental identities—universality $P' = \sum Q(\alpha_j, \cdot)$, the Turán SOS $P'^2 - PP'' = \sum Q_j^2$, and the quotient decomposition $P = (x-a)Q + P(a)$—all hold over $\mathbb{F}_p$. We then interpret Dvir's polynomial method proof of the Kakeya conjecture through the Pandrosion fiber: the restriction of a vanishing polynomial to each line yields fiber quotients $Q(a,t)$ whose degree constraints force the polynomial to be identically zero, yielding $|K| \geq C(n,q) \cdot q^n$.

1 Pandrosion arithmetic over $\mathbb{F}_p$

Theorem 1.1 (Pandrosion universality over $\mathbb{F}_p$). *For $P \in \mathbb{F}_p[x]$ with roots $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_p$ (fully split), the Pandrosion quotient $Q(\alpha_j, x) = P(x)/(x-\alpha_j) \in \mathbb{F}_p[x]$ satisfies:*

$$P'(x) \equiv \sum_{j=1}^{n} Q(\alpha_j, x) \pmod{p}. \tag{1}$$

Theorem 1.2 (Fermat–Pandrosion polynomial). *Over $\mathbb{F}_p$:*

$$x^p - x = \prod_{a \in \mathbb{F}_p} (x-a). \tag{2}$$

The Pandrosion field of $x^p - x$ is $F(x) = (px^{p-1} - 1)/(x^p - x) \equiv -1/(x^p - x) \pmod{p}$.

Theorem 1.3 (Turán SOS over $\mathbb{F}_p$). *For fully split $P \in \mathbb{F}_p[x]$:*

$$P'(x)^2 - P(x)P''(x) \equiv \sum_{j=1}^{n} Q(\alpha_j, x)^2 \pmod{p}. \tag{3}$$

Verified for $p = 5, 7, 11, 13$ with 0 violations across 166 tests of fully split polynomials.

2 Dvir's theorem via Pandrosion fibers

Definition 2.1. A *Kakeya set* $K \subseteq \mathbb{F}_q^n$ is a set containing a line in every direction: for each $b \in \mathbb{F}_q^n \setminus \{0\}$, there exists a such that $\{a + tb : t \in \mathbb{F}_q\} \subseteq K$.

Theorem 2.2 (Dvir, 2008).

$$|K| \geq \binom{n+q-1}{n} \geq \frac{q^n}{n!}. \tag{4}$$

Pandrosion proof. Assume $|K| < \binom{n+q-1}{n}$.

Step 1. Since $|K| < \dim(\mathbb{F}_q[x_1, \ldots, x_n]_{\leq q-1})$, there exists a nonzero polynomial P of degree $d \leq q-1$ vanishing on K.

Step 2 (Pandrosion restriction). For each direction b, the line $\ell_b = \{a+tb : t \in \mathbb{F}_q\} \subseteq K$. Then $P|_{\ell_b}(t) = P(a+tb)$ vanishes for all $t \in \mathbb{F}_q$. Since $\deg P|_{\ell_b} \leq d \leq q-1 < q$, a polynomial of degree $< q$ with q roots must be identically zero.

In Pandrosion language: $t^q - t = \prod_{s \in \mathbb{F}_q}(t-s)$ annihilates $P|_{\ell_b}$. The quotient $Q(s,t) = P|_{\ell_b}/(t-s)$ at each s-fiber contributes to the universality $P|'_{\ell_b} = \sum_s Q(s,t)$, but since $P|_{\ell_b} \equiv 0$, every fiber vanishes.

Step 3 (Leading form). The leading homogeneous part $P_d(x)$ satisfies: $P_d(b) = 0$ for *every* direction $b \in \mathbb{F}_q^n$. Over $\mathbb{F}_q$, this gives $q^n - 1$ vanishing points for a polynomial of degree $d < q$, which forces $P_d \equiv 0$, contradicting $\deg P = d$. □

3 Kakeya sets: numerical bounds

$\mathbb{F}_q^2$	$\lvert K \rvert$ (random)	q^2	Ratio
$\mathbb{F}_3^2$	7	9	0.78
$\mathbb{F}_5^2$	19	25	0.76
$\mathbb{F}_7^2$	35	49	0.71

The random Kakeya sets use about 3/4 of the full field—far above Dvir's lower bound of $\sim q^2/2$.

4 Verification

Identity (mod p)	**Tests**	**Violations**
$x^p - x = \prod(x-a)$	$p = 3, 5, 7, 11$	0
$P = (x-a)Q + P(a)$	3×200	0
$P' = \sum Q(\alpha_j, \cdot)$	3×100	0
$P'^2 - PP'' = \sum Q^2 \pmod{p}$	166	0

References

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[3] I. Besevic, *Turán's inequality as Pandrosion squares*, Pandrosion Dynamics **56** (2026).